INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Introduction to Cryptography

Lab Session 01: Affine cipher *February 19, 2026*

Please solve the following exercises in pairs. You can use C/C++, C#, Java or Python do develop your source code, but use only one of them to do all the exercises.

1. Programming exercises

Use the alphabet of the printable characters in the ASCII i.e. $A = \{space, '!', \ldots, \tilde{}\ \}$ with codes from 32 to 126. Keep the line breaks of the original plaintext. Then do the following:

1. Design and implement your own function to compute the greatest common divisor of two integers a, b
2. Design and implement a function that receives as a parameter an integer $n \geq 2$. Your function must create a list with the elements of $\mathbb{Z}_n^*$. Use the function you implemented for gcd (point 1)
3. Design and implement a function that receives as parameters an integer $n \geq 2$ and an element $a \in \mathbb{Z}_n^*$, your function must test every element in $\mathbb{Z}_n^*$ to find and return the element $b \in \mathbb{Z}_n^*$, such that $ab \text{ mód } n = 1$.
4. Use the alphabet of the printable characters in the ASCII i.e. $A = \{space, '!', \ldots, \tilde{}\ \}$ with codes from 32 to 126 and to do the following:
   a) Design and implement a key generation function considering that the alphabet is the printable ASCII.Your function must randomly generate a valid key $K = (a, b)$ for the affine cipher and print it. Use the functions that you designed for points 2 and 3.
   b) Design and implement a function to encipher a text file using the affine cipher. Your function must receive as a parameter the filename of the plaintext, a valid key K for the affine cipher, store the ciphertext in a file. Your function must use modular arithmetic with integers module n, where n is the size of the alphabet.
   c) Design and implement a function to decipher a ciphertext enciphered with the affine cipher. Your function must receive a ciphertext (with characters in the set A), a valid key K for the affine cipher, and recover the original plaintext. Use the function of point 3 to find $a^{-1} \text{ mód } n$. Your function must use modular arithmetic with integers module n, where n is the size of the alphabet.

5. You must test your implementation in pairs as follows :

   a) One student of the team, must generate at random a valid key $K$for the affine cipher considering that the alphabet is the printable ASCII. Then he/she must encipher a text file, for example the source code of any program, using the generated key K. **The filename of the plaintext and the ciphertext, and the key must be given by the user. You must not need to write the filename or the key directly into the source code.** Once he/she obtained the ciphertext, the student must share the key K and the textfile of the ciphertext with his/her partner.

   b) The other student must use K and the textfile of the ciphertext C to recover the plaintext M. **The filename of the ciphertext, the recovered plaintext and the key must be given by the user. You must not need to write the filename or the key directlyinto the source code.**. Your program must print the value of a^{-1} mód n.

2. Questions

Answer the following questions and write the answers in your report

1. What is the mathematical expression to compute $\varphi(n)$ the Euler phi function ? What does it count?
2. Compute the value of the Euler phi function φ for $n = 101$; $n = 35$, $n = 101$, $n = 10403$, $n = 4375$, $n = 101$, $n = 225$, $n = 101$ $n = 385$. Write the details of your calculation.
3. How can we use the Euler phi function to compute the number of valid keys for the affine cipher? What is the specific value of valid keys for the alphabet of printable ascii characters?
4. Write down the pseudocode of an algorithm that receives as input a ciphertext C and the size n of the alphabet used. The algorithm must try to find the corresponding plaintext M. Use a small example to explain how your algorithm works?
5. Write down the reference you read to answer the questions.

3. Products

Every student must write her/his own report, containing at least:

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. List 10 keys obtained by your key generation function.
3. Give 5 pairs of plaintext/ciphertext you used to test your enciphering function.
4. Answers to the questions of Section 2.